Richmond Valley Council

Richmond Valley Development Control Plan 2021 (Transitional Flood Provision) Amendment- Transitional Flood Planning Arrangements post adoption of the 2026 Richmond Valley Flood Risk Management Plan

August 2026

1. Citation

This Plan may be cited as *Richmond Valley Development Control Plan 2021 (Transitional Flood Provision) Amendment*. It constitutes an amendment to the *Richmond Valley Development Control Plan 2021 (as amended)* as provided by Section 3.43(4) of the *Environmental Planning and Assessment Act 1979*.

2. Commencement

This Plan commences on the day following Council's endorsement of the *Richmond Valley Council Flood Risk Management Study & Plan (2026)*.

3. Land to which this Plan applies

This Plan applies to all land within the Richmond Valley Council local government area.

4. Relationship to other Planning Instruments

This Plan provides for the amendment of *Richmond Valley Development Control Plan 2021 (as amended)*.

5. Aims of this Plan

The aim of this Plan is to amend the *Richmond Valley Development Control Plan 2021 (as amended)* to adopt transitional flood planning control arrangements while detailed DCP amendments are prepared in response to recommendations from the *Richmond Valley Flood Risk Management Study & Plan (2026)*.

6. Savings and Transitional

This DCP Amendment does not apply to an application under the *Environmental Planning and Assessment Act 1979* which was lodged with Council but not finally determined before the commencement of this DCP Amendment. Any application lodged before this amendment commenced will be assessed in accordance with the relevant DCP provision which applied at the time of application lodgement.

7. Amendment of Richmond Valley Council Development Control Plan 2021

[1] Introduction

Delete from Section 2 *Commencement* "This amendment to the DCP was approved by Richmond Valley Council on 21 July 2026 and becomes effective from 27 July 2026."

[2] Introduction

Insert the following row into the table to Section 9 *Amendments to this plan-*

2	Transitional Flood Planning Provision	18 August 2026	19 August 2026

[3] Part H-1 Flood Planning

Replace the chapter with the attached chapter.

Richmond Valley Development Control Plan 2021

Part H-1

Flood Planning

This DCP applies to all land within the Richmond Valley Local Government Area.

Date adopted by Council:
22/06/2021

Effective Date:
01/08/2021

Amendments:
Amendment 1 – Effective 27 July 2026
Amendment 2 – Effective 19 August 2026

This Chapter provides guidance for development of Flood Prone Land and should be read in conjunction with the *NSW Flood Risk Management Manual* and Council's adopted *Floodplain Risk Management Plan*.

H-1.1 General Objectives

The general objectives of this Chapter are to:

(1) align flood planning with the NSW Government's Flood Prone Land Policy and the NSW Flood Risk Management Manual.

(2) explain the relevance of the adopted Flood Planning Area.

(3) establish transitional arrangements following the adoption of the *Richmond Valley Flood Risk Management Study & Plan 2026*.

(4) explain the floodplain risk hazard categories and Flood Planning Constraint Categories (FPCC), and encourage suitable development compatible with flood hazard.

(5) adopt a risk-based approach to addressing flooding in planning decisions.

H-1.2 Flood Risk Management

Objectives

(a) to explain the flood risk categories, including FPCC, adopted by Council's *Flood Risk Management Plan*.

(b) to recognise the 1% AEP (or 1 in 100-year ARI) design flood (plus climate change and a 500mm freeboard) as the Defined Flood Event (DFE).

(c) shift from referencing Average Recurrence Interval (ARI) to Annual Exceedance Probability (AEP).

(d) embrace the Flood Planning Constraint Category (FPCC) framework for testing site suitability while retaining the Flood Planning Control matrices from the Floodplain Risk Management Plans for Casino and the Mid-Richmond to establish flood planning controls.

Controls

(1) The *Richmond Valley Flood Risk Management Plan 2026* (the new Plan) was adopted by Council on 18 August 2026 and replaced two *Floodplain Risk Management Plans* (the former Plans), one for each of Casino and the Mid-Richmond.

(1A) As a transitional arrangement, the Flood Planning matrices from the former Plans will continue to be used by this DCP for development controls. This transitional arrangement will continue until new controls, and a DCP amendment, can be developed per recommendations from the new Plan.

The scope of the former Flood Planning matrices will also be extended to apply to the entire floodplain of the LGA.

Note. Where there is doubt over which matrix will apply to land, Council will decide based upon the land's location in the catchment (ie upper catchment verse lower catchment) and the relevance of the respective former Plan to the land.

(2) The *NSW Flood Risk Management Manual* advocates a risk-based approach when addressing flooding in planning decisions. This includes-

- the level of assessment undertaken for a development proposal is proportionate to the likely impacts of the proposal, including taking into account the relative flood risk of the proposal.
- planning decisions are based upon a balanced consideration of the merits, risks and impacts of a given proposal, and that appropriate measures are in place to limit impacts to an acceptable level and achieve a tolerable flood risk level for flood-affected proposals.

(3) With few exceptions, it is recognised that it is neither feasible nor socially or economically justifiable to adopt the PMF as the basis for flood planning. Council's new Plan adopts the 1%AEP (with climate change and a 500mm freeboard) flood event as the Defined Flood Event (DFE). This is the benchmark for managing flood risk in land-use planning and development decisions.

Note. A 1% AEP flood may also be referred to as a 1 in 100-year ARI flood event —measured as having a 1% probability of occurring or being exceeded in any single year.

(4) The former Risk Plans contained reference to Floodplain Hazard Categories. These are tools for assessing the suitability and minimum requirements for development in the Flood Planning Control Matrices. They are based on a combination of depth (D) and velocity (V). These categories are:

(a) **High Floodway Hazard (HFH)** - based on a 1%AEP (with climate change) (or 1 in 100 year) design flood – Flow paths that carry significant volumes of flood water during a 100-year flood. Danger to life and limb, evacuation difficult, potential for structural damage, high social disruption, and economic losses.
V>2m/s or VxD>1 [for D>1m] or D+(0.3xV)>1 [for V>1m/s]

(b) **High Depth Hazard (HDH)** - based on a 1%AEP (with climate change) (or 1 in 100 year) design flood – Area where floodwaters are deep but are not flowing with high velocity.
V<1m/s and VxD<1 or D+(0.3xV)>1

(c) **High Isolation Hazard (HIH)** - based on a 1%AEP (with climate change) (or 1 in 100 year) design flood – As per High Depth but with no easy access to safe refuge (ie more than 500m to high ground)

(d) **Possible High Depth Hazard (HFH) or Low Hazard (LH)** - based on a 1%AEP (with climate change) (or 1 in 100 year) design flood – Insufficient ground level information. Final category dependent on the exact ground levels at the particular site.

(e) **Low Hazard (LH)** - based on a 1%AEP (with climate change) (or 1 in 100 year) design flood – Flood depths and velocities are sufficiently low that people and their possessions can be evacuated.

(f) **Rare Low Hazard (RLH)** - based on PMF - Any land that is inundated in the PMF event and has not been assigned one of the other hazard categories. These areas are generally above the 1%AEP (with climate change) (or 1 in 100 year) design flood.

(g) **Rare High Floodway Hazard (RHFH)** - based on 0.2%AEP (with climate change) (or 1 in 500 year) design flood - Flow paths that carry significant volumes of flood water during a 500-year design flood.

These rare event areas may present danger to life and limb, create evacuation difficulties, and increase potential for structural damage, high social disruption, and economic losses.

(5) The new Plan adopts the Flood Planning Constraint Category (FPCC) framework to classify land within the floodplain according to flood extent, hazard, flood function, flood planning areas and emergency response considerations, across a range of flood frequencies. The purpose of this framework is to distinguish areas suitable for more intensive development from those where constraints are significant. The four FPCCs adopted are-

(a) FPCC1—extreme flood hazard—includes floodways and key flood storage areas in the DFE, and areas subject to H6 hazard. These are the most constrained areas where most new or intensified development is generally unsuitable.

(b) FPCC2—high flood hazard—includes floodways and H6 hazard areas in events larger than the DFE, H5 hazard areas in the DFE, and locations with emergency response constraints. Development may be possible but requires detailed assessment and careful management of risk.

(c) FPCC3—moderate flood hazard—comprises the remaining land within the Flood Planning Area below the DFE plus freeboard. These areas are generally more suitable for urban infill or redevelopment, subject to appropriate flood related controls.

(d) FPCC4—low flood hazard—includes land between the Flood Planning Area and the Probable Maximum Flood. While hazard is lower, flood related planning considerations may still apply, particularly for critical infrastructure.

Map_Set_O_Flood_Planning_Constraint_Categories_With_Relaxed_Flood_Storage_Criteria from Volume 2 of the new Plan is adopted by this DCP for the purposes of testing site suitability, see Figure H-1.

Note. This mapping is available on Council's Intramaps Spatial Online Viewer at https://richmondvalley.nsw.gov.au/building-and-planning-services/planning-instruments-and-guidelines/ .

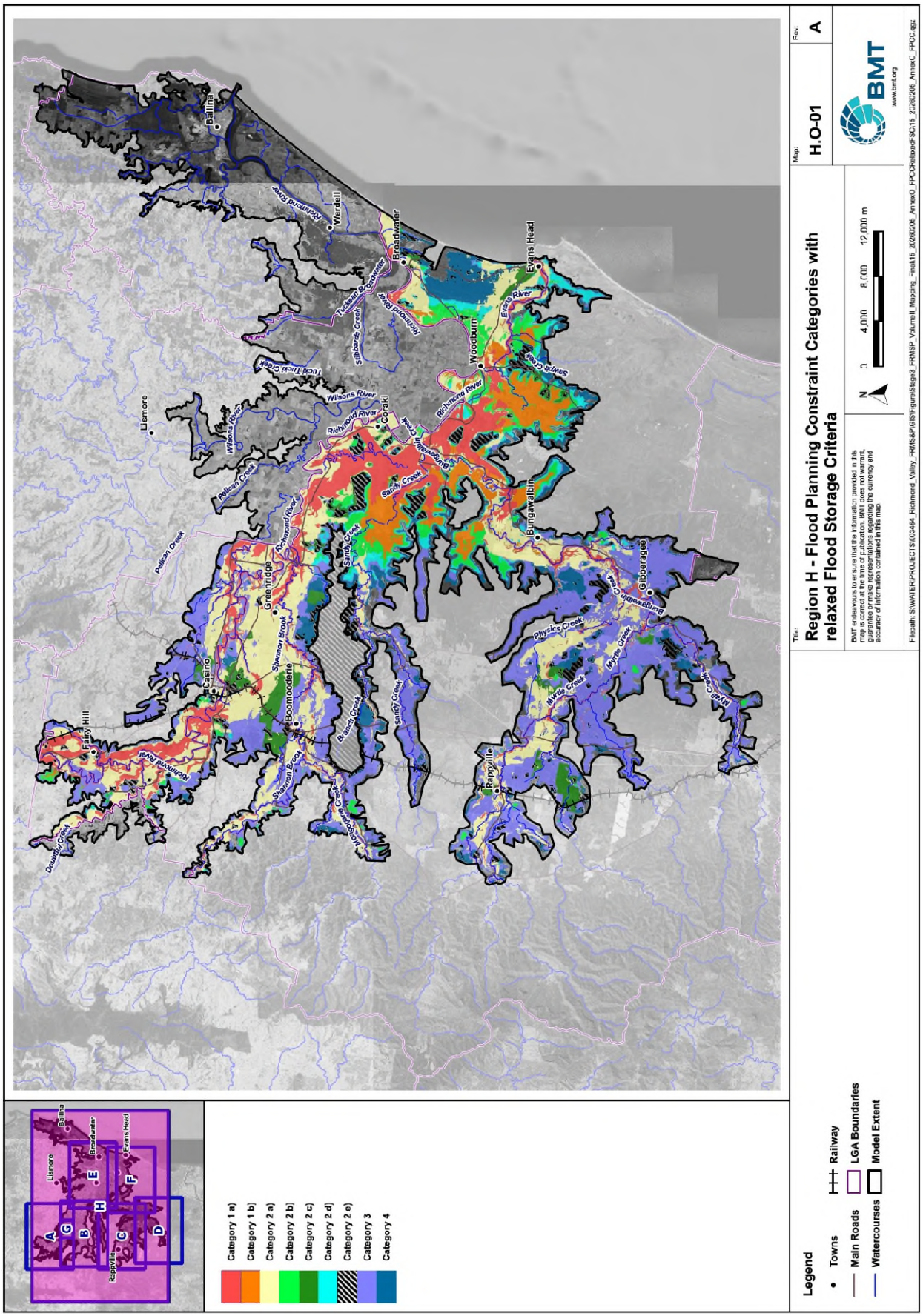


Figure H-1 – FPCC Map (extracted from the Richmond Valley Flood Risk Management Study & Plan (2026) Volume 2 - Map Set O – Flood Planning Constraints Categories with relaxed Flood Storage Criteria)

H-1.3 Flood Planning Area v Flood Planning Level

Objectives

(a) To explain what is meant by the Flood Planning Area (FPA) and Flood Planning Level (FPL).

Controls

(1) Council's *Flood Risk Management Plan* adopts the 1%AEP (with climate change and a 500mm freeboard) as the Defined Flood Event (DFE). The DFE underpins key flood risk management outputs, including the Flood Planning Area (FPA), Flood Function, Flood Hazard, Flood Planning Constraints, and related mapping and assessments.

Note. Clause 5.21 of the LEP applies to land below the Flood Planning Area (FPA). This is taken to be the DFE.

Clause 5.22 of the LEP applies to land above the FPA and below the PMF.

(2) The Flood Planning Level (FPL) can vary from location to location or for development type to development type. It is used to establish minimum floor level requirements.

For residential development, the FPL adopted by the new Plan and former Plans are the same as the DFE, that is the 1%AEP (with climate change + 500mm freeboard).

For industrial/commercial development, the former Plans adopt different FPL standards. The Mid-Richmond Plan requires a minimum floor level at the 5%AEP (with climate change) (or 1 in 20-year ARI), while the Casino Plan requires a minimum floor level at the 1%AEP (with climate change) (or 1 in 100 yr ARI). These former FPLs will continue to apply during the transitional period.

H-1.4 Transitional Flood Planning Controls for development

Objectives

(a) To retain flood planning controls from the former *Floodplain Risk Management Plans* as a transitional arrangement until recommendations from the new Plan, including development of a FPCC Planning Control Matrix and DCP Amendment, can be approved.

(b) Allow flexibility in flood planning controls, without compromising the safety of residents and the community.

(c) To apply planning controls from the former Plans to areas of flood plain outside their original scope.

Controls

(1) Council's *Floodplain Risk Management Plans* (the former Plans) adopt various flood planning control requirements. The former Plans should be the primary source of appropriate flood planning controls during the transitional period. Copies of the Flood Planning Matrices are attached to this chapter.

(2) **Residential development**

(a) The residential Flood Planning Level shall be the 1%AEP plus climate change and a freeboard of 500mm.

(b) The floor level of habitable rooms is to be erected above the residential Flood Planning Level (FPL).

(c) Some exceptions to (b) will be permitted for minor extensions to existing dwellings.

(3) **Commercial & Industrial Development**

(a) The commercial/industrial Flood Planning Level shall be for the-

(i) Mid Richmond – the 5%AEP (plus climate change) (or 1 in 20-year ARI) flood event. Storage areas with a flood level greater than the 1%AEP (with climate change) flood event is to be provided for stock/equipment susceptible to flood damage.

(ii) Casino – the 1%AEP (plus climate change) (or 1 in 100-year) flood event.

(b) Development which includes storage of hazardous materials must store these materials above the 1%AEP (plus climate change) flood event plus an appropriate Risk-Based Freeboard (RBF), or as per an approved plan of management.

(c) A combination of design, floor level and freeboard will be used to determine the suitability of development through consultation of the Risk Plans.

(4) **Other Development**

(a) A combination of design, floor level and freeboard will be used to determine the suitability of development through consultation of the Risk Plans and Flood Planning Matrices.

H-1.5 Flood information

Objectives

(a) To ensure that flood information is readily available to the community.

(b) To establish what flood information may be relied upon for areas outside the Richmond Valley Flood Study 2023.

Design Criteria

(1) Flood information relevant to individual properties, based upon contemporary design flood modelling, is available from Council for land within the extent of the Richmond Valley Flood Study.

(2) For localities outside a modelled area, the proponent of a development may be required to predict the flood planning level by conducting a localised flood assessment, including utilising anecdotal evidence of past flood heights and consequences.

A registered Surveyor must be used to establish the level of flooding over the land and issue a Compliance Certificate along with details of the source of the

flood information, and the height of natural ground level, the flood level and finished floor level for the building site. In addition, a diagram shall be provided identifying the location of the building site and any other relevant information such as location of benchmarks used to source the flood information.

Note. A Flood Information Report containing information relevant to a parcel of land can be obtained from Council for a fee. Applicants seeking development consent on flood liable land are encouraged to obtain this flood information to support their applications. Reports can be obtained via this link –

https://richmondvalley.nsw.gov.au/services/flood/flood-information-report-enquiry/

CASINO FLOOD PLANNING MATRIX

TABLE 1: RESIDENTIAL, COMMERCIAL AND INDUSTRIAL DEVELOPMENT WITHIN AN URBAN AREA

Controls	Development / Building Type	No Hazard	Flood Hazard Category					Additional Constraint[1]
			Rare Low Hazard[2]	Low Hazard	High Depth Hazard	High Isolation Hazard	High Floodway Hazard	Rare High Floodway Hazard[2]
Land Use Suitability & Fill Level	Existing Lot - New Development	N/A	SF1	SF1	SF1	SF1		SF1
	(this line not used)							
	Subdivision	N/A	SF2	SF2	SF2			
	Emergency Services Site (Hospitals, etc.)	N/A	SF3a	SF3a				
	Other Community Service Building (School, etc.)	N/A	SF3b	SF3b				
Floor Level	New Habitable Building	N/A	FL2c	FL2c	FL2c	FL2c		FL2c
	New Commercial or Industrial Building	N/A	FL2a	FL2a	FL2a	FL2a		FL2a
	New Emergency Service Building (Hospitals, etc.)	FL3a	FL3a	FL3a				
	New Other Community Service Building (School, etc.)	FL3b	FL3b	FL3b				
	New Ancillary Building (eg shed, carport)	N/A	FL1	FL1	FL1	FL1		FL1
	Building Extension	N/A	FL4a	FL4a	FL4b	FL4b		FL4b
Building Components		N/A	BC1	BC1	BC1	BC1		BC1
Structural Soundness	Ancillary Building (eg. shed, carport)	N/A	SS1	SS1	SS1	SS1		SS2
	Other Building	N/A	SS1	SS1	SS2	SS2		SS3
Flood Effect	Existing Lot - New Development	N/A	FE1	FE2	FE2	FE2		FE2
	Subdivision	N/A	FE2	FE2	FE2			FE3
	New Ancillary Building (eg shed, carport)	N/A	FE1	FE2	FE2	FE2		FE2
	Building Extension	N/A	FE1	FE1	FE2	FE2		FE3
	Other Developments (road raising, etc)	N/A	FE1	FE2	FE2	FE2	FE3	FE3
Evacuation & Access	Existing Lot - New Development	N/A	EA1	EA1	EA1	EA1		EA1
	Subdivision	N/A	EA3	EA3	EA3			
	Emergency Service Site (Hospitals, etc.)	N/A	EA4a	EA4a				
	Other Community Service Site (Schools, etc.)	N/A	EA4b	EA4b				
Flood Awareness, etc		N/A	FA2	FA2	FA2	FA2	FA2	FA2

Note 1: In addition to being assigned one of the standard flood hazard categories, a site may be classified as a "Rare High Floodway Hazard". In this instance, the most stringent of the two controls is to be used. For example, if the site is classified as both "Low Hazard" and "Rare High Floodway Hazard", no community services buildings are permitted (because "unsuitable" is more stringent than SF3b)

Note 2: The extreme flood hazard categories (i.e. "Rare Low Hazard" and "Rare High Floodway Hazard") are applicable only to the 2D model region in Casino An explanation of the criteria used to define the hazard categories is contained in the Casino Floodplain Management Study (WBM Oceanics Australia, 2001)

Control Measures

N/A	Controls Not Applicable
	Unsuitable Land Use - Not considered suitable for development
	LAND USE SUITABILITY & MINIMUM FILL LEVEL
SF1	Consider for development subject to the controls below. No minimum fill level required.
SF2	Consider for development subject to the controls below. For residential and commercial areas, the minimum fill level to be greater than or equal to the 100 year flood level. For industrial areas, the minimum fill level to be greater than or equal to the 10 year flood level.
SF3a	Consider for development subject to the controls below. Minimum fill level greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
SF3b	Consider for development subject to the controls below. Council to give consideration on the benefits of using the development during and after a flood emergency. If the site is to be used for a flood emergency, the minimum fill level should preferably be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
	MINIMUM FLOOR LEVEL
FL1	No minimum floor level required (Council to advise developer of flood risk and potential damage to building & contents. Flood levels available on request)
FL2a	All floor levels to be greater than or equal to the 100 year flood level
FL2b	Not used
FL2c	All floor levels to be greater than or equal to the 100 year flood level plus 0.5m
FL3a	All floor levels to be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL3b	If practical, some or all floor levels to be greater than or equal to the PMF flood level, so that these buildings will be available for accommodation / storage during and after a flood emergency. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL4a	All floor levels to be as close to the *minimum floor level* above (habitable or other) as practical and not less than the floor level of the existing building being extended if the existing floor level is less than or equal to the minimum floor level. If the extended weatherproof area exceeds 50% of the existing weatherproof area, the extension is treated as a new building. The extended weatherproof area is measured as the cumulative area of any previous extensions plus the proposed extension.
FL4b	As for FL4a with the maximum percentage increase in extended weatherproof area to be: (a) 50% if the extension's floor level is less than one (1) metre below the 100 year flood level; (b) 25% if the extension's floor level is greater than two (2) metres below the 100 year flood level; or (c) pro-rata between 50% and 25% for floor levels from one (1) metre to two (2) metres below the 100 year flood level.
	BUILDING COMPONENTS
BC1	Buildings to have flood compatible material below the higher of (a) the minimum floor level or (b) the 1 in 100year flood level plus 0.5m.
	STRUCTURAL SOUNDNESS
SS1	No structural soundness requirements for the force of floodwater, debris & buoyancy
SS2	Engineers report to prove that structures subject to a flood up to the 100 year event can withstand the force of floodwater, debris & buoyancy.
SS3	Engineers report to prove that structures subject to a flood up to the 500 year event can withstand the force of floodwater, debris & buoyancy.
	FLOOD EFFECT
FE1	No action required
FE2	The flood impact of the development to be considered by Council, with Council having the right to request an engineer's report (see FE3 below)
FE3	Engineers report required to prove that the development will not result in adverse flood impact elsewhere
	EVACUATION/ACCESS
EA1	Council to provide information on flood evacuation strategy
EA2	Not used
EA3	Reliable access for pedestrians and transport required during the 100yr ARI event. Council to provide information on flood evacuation strategy
EA4a	Emergency service site - should have good access up to the PMF and preferably not cut-off from the main residential area(s). Council to evaluate suitablility of site in this respect.
EA4b	If site to be used during and after a flood emergency (see FL3b above), should have good access up to the PMF and preferably not cut-off from the main residential area(s).
	FLOOD AWARENESS
FA1	Not used
FA2	S149(2) Certificates to notify possible affectation by a flood in the Richmond River and/or one of its tributaries. The severity of flooding can be determined by comparison of surveyed levels of the site with predicted flood heights, and also the flood hazard.

CASINO FLOOD PLANNING MATRIX

TABLE 2: DEVELOPMENT IN RURAL AREAS

Controls	Development / Building Type	No Hazard	Flood Hazard Category: Rare Low Hazard[2]	Low Hazard	High Depth Hazard	High Isolation Hazard	High Floodway Hazard	Additional Constraint[1]: Rare High Floodway Hazard[2]
Land Use Suitability & Fill Level	Habitable Building	N/A	SF1	SF1	SF1	SF1		SF1
	Ancillary Building (eg. shed)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	Other Developments (eg. levees, roads, dams, etc)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	Emergency Services Site (Hospitals, etc.)	N/A	SF3a	SF3a				
	Other Community Service Building (School, etc.)	N/A	SF3b	SF3b	SF3b			
Floor Level	New Habitable Building	N/A	FL2c	FL2c	FL2c	FL2c		FL2c
	(this line not used)							
	New Emergency Service Building (Hospitals, etc.)	FL3a	FL3a	FL3a				
	New Other Community Service Building (School, etc.)	FL3b	FL3b	FL3b				
	New Ancillary Building (eg shed, carport)	N/A	FL1	FL1	FL1	FL1	FL1	FL1
	Building Extension	N/A	FL4a	FL4a	FL4b	FL4b		FL4b
Building Components		N/A	BC1	BC1	BC1	BC1	BC1	BC1
Structural Soundness	Small-scale[3] Development (eg. shed, small dam)	N/A	SS1	SS1	SS1	SS1	SS2	SS3
	Large-scale Development (eg. levee, raised road)	N/A	SS1	SS1	SS2	SS2	SS2	SS3
Flood Effect	Small-scale[3] Development (eg. shed, small dam)	N/A	FE1	FE1	FE2	FE2	FE2	FE2
	Large-scale Development (eg. levee, raised road)	N/A	FE2	FE2	FE3	FE3		FE3
	(this line not used)							
	(this line not used)							
	(this line not used)							
Evacuation & Access	Habitable Building	N/A	EA1	EA1	EA1	EA1		EA1
	(this line not used)							
	Emergency Service Site (Hospitals, etc.)	N/A	EA4a	EA4a				
	Other Community Service Site (Schools, etc.)	N/A	EA4b	EA4b				
Flood Awareness, etc		N/A	FA2	FA2	FA2	FA2	FA2	FA2

Note 1: In addition to being assigned one of the standard flood hazard categories, a site may be classified as a "Rare High Floodway Hazard". In this instance, the most stringent of the two controls is to be used. For example, if the site is classified as both "Low Hazard" and "Rare High Floodway Hazard", no community services buildings are permitted (because "unsuitable" is more stringent than SF3b)

Note 2: The extreme flood hazard categories (i.e. "Rare Low Hazard" and "Rare High Floodway Hazard") are applicable only to the 2D model region in Casino An explanation of the criteria used to define the hazard categories is contained in the Casino Floodplain Management Study (WBM Oceanics Australia, 2001)

Note 3: Small-scale development implies development on rural land that is small relative to the width of the floodplain and is not part of a planned large-scale development.

Control Measures

N/A	Controls Not Applicable
	Unsuitable Land Use - Not considered suitable for development
	LAND USE SUITABILITY & MINIMUM FILL LEVEL
SF1	Consider for development subject to the controls below. No minimum fill level required.
SF2	Consider for development subject to the controls below. For residential and commercial areas, the minimum fill level to be greater than or equal to the 100 year flood level. For industrial areas, the minimum fill level to be greater than or equal to the 10 year flood level.
SF3a	Consider for development subject to the controls below. Minimum fill level greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
SF3b	Consider for development subject to the controls below. Council to give consideration on the benefits of using the development during and after a flood emergency. If the site is to be used for a flood emergency, the minimum fill level should preferably be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
	MINIMUM FLOOR LEVEL
FL1	No minimum floor level required (Council to advise developer of flood risk and potential damage to building & contents. Flood levels available on request)
FL2a	All floor levels to be greater than or equal to the 100 year flood level
FL2b	Not used
FL2c	All floor levels to be greater than or equal to the 100 year flood level plus 0.5m
FL3a	All floor levels to be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL3b	If practical, some or all floor levels to be greater than or equal to the PMF flood level, so that these buildings will be available for accommodation / storage during and after a flood emergency. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL4a	All floor levels to be as close to the *minimum floor level* above (habitable or other) as practical and not less than the floor level of the existing building being extended if the existing floor level is less than or equal to the minimum floor level. If the extended weatherproof area exceeds 50% of the existing weatherproof area, the extension is treated as a new building. The extended weatherproof area is measured as the cumulative area of any previous extensions plus the proposed extension.
FL4b	As for FL4a with the maximum percentage increase in extended weatherproof area to be: (a) 50% if the extension's floor level is less than one (1) metre below the 100 year flood level; (b) 25% if the extension's floor level is greater than two (2) metres below the 100 year flood level; or (c) pro-rata between 50% and 25% for floor levels from one (1) metre to two (2) metres below the 100 year flood level.
	BUILDING COMPONENTS
BC1	Buildings to have flood compatible material below the higher of (a) the minimum floor level or (b) the 1 in 100year flood level plus 0.5m.
	STRUCTURAL SOUNDNESS
SS1	No structural soundness requirements for the force of floodwater, debris & buoyancy
SS2	Engineers report to prove that structures subject to a flood up to the 100 year event can withstand the force of floodwater, debris & buoyancy.
SS3	Engineers report to prove that structures subject to a flood up to the 500 year event can withstand the force of floodwater, debris & buoyancy.
	FLOOD EFFECT
FE1	No action required
FE2	The flood impact of the development to be considered by Council, with Council having the right to request an engineer's report (see FE3 below)
FE3	Engineers report required to prove that the development will not result in adverse flood impact elsewhere
	EVACUATION/ACCESS
EA1	Council to provide information on flood evacuation strategy
EA2	Not used
EA3	Reliable access for pedestrians and transport required during the 100yr ARI event. Council to provide information on flood evacuation strategy
EA4a	Emergency service site - should have good access up to the PMF and preferably not cut-off from the main residential area(s). Council to evaluate suitablility of site in this respect.
EA4b	If site to be used during and after a flood emergency (see FL3b above), should have good access up to the PMF and preferably not cut-off from the main residential area(s).
	FLOOD AWARENESS
FA1	Not used
FA2	S149(2) Certificates to notify possible affectation by a flood in the Richmond River and/or one of its tributaries. The severity of flooding can be determined by comparison of surveyed levels of the site with predicted flood heights, and also the flood hazard.

CASINO FLOOD PLANNING MATRIX

TABLE 3: OTHER

Controls	Development / Building Type	No Hazard	Rare Low Hazard[2]	Low Hazard	High Depth Hazard	High Isolation Hazard	High Floodway Hazard	Rare High Floodway Hazard[2]
			Flood Hazard Category					Additional Constraint[1]
Land Use Suitability & Fill Level	Non-Habitable Building (shed, toilets, shelter, etc)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	(this line not used)							
	Other Developments (eg. levees, roads, dams, etc)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	(this line not used)							
	(this line not used)							
Floor Level	(this line not used)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	(this line not used)							
	(this line not used)							
	(this line not used)							
	(this line not used)							
	(this line not used)							
Building Components		N/A	BC1	BC1	BC1	BC1	BC1	BC1
Structural Soundness	Small-scale[3] Development (eg. shed, small dam)	N/A	SS1	SS1	SS1	SS1	SS2	SS3
	Large-scale Development (eg. levee, raised road)	N/A	SS1	SS1	SS2	SS2	SS2	SS3
Flood Effect	Small-scale Development (eg. shed, small dam)	N/A	FE1	FE1	FE2	FE2	FE2	FE2
	Large-scale Development (eg. levee, raised road)	N/A	FE2	FE2	FE3	FE3	FE3	FE3
	(this line not used)							
	(this line not used)							
	(this line not used)							
Evacuation & Access	Not Applicable							
	(this line not used)							
	(this line not used)							
	(this line not used)							
Flood Awaren	Not Applicable							

Note 1: In addition to being assigned one of the standard flood hazard categories, a site may be classified as a "Rare High Floodway Hazard". In this instance, the most stringent of the two controls is to be used. For example, if the site is classified as both "High Depth Hazard" and "Rare High Floodway Hazard", an engineer's report is required (because SS3 is more stringent than SS2)

Note 2: The extreme flood hazard categories (i.e. "Rare Low Hazard" and "Rare High Floodway Hazard") are applicable only to the 2D model region in Casino
An explanation of the criteria used to define the hazard categories is contained in the Casino Floodplain Management Study (WBM Oceanics Australia, 2001)

Control Measures

N/A	Controls Not Applicable
	Unsuitable Land Use - Not considered suitable for development
	LAND USE SUITABILITY & MINIMUM FILL LEVEL
SF1	Consider for development subject to the controls below. No minimum fill level required.
SF2	Consider for development subject to the controls below. For residential and commercial areas, the minimum fill level to be greater than or equal to the 100 year flood level. For industrial areas, the minimum fill level to be greater than or equal to the 10 year flood level.
SF3a	Consider for development subject to the controls below. Minimum fill level greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
SF3b	Consider for development subject to the controls below. Council to give consideration on the benefits of using the development during and after a flood emergency. If the site is to be used for a flood emergency, the minimum fill level should preferably be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
	MINIMUM FLOOR LEVEL
FL1	No minimum floor level required (Council to advise developer of flood risk and potential damage to building & contents. Flood levels available on request)
FL2a	All floor levels to be greater than or equal to the 100 year flood level
FL2b	Not used
FL2c	All floor levels to be greater than or equal to the 100 year flood level plus 0.5m
FL3a	All floor levels to be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL3b	If practical, some or all floor levels to be greater than or equal to the PMF flood level, so that these buildings will be available for accommodation / storage during and after a flood emergency. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL4a	All floor levels to be as close to the *minimum floor level* above (habitable or other) as practical and not less than the floor level of the existing building being extended if the existing floor level is less than or equal to the minimum floor level. If the extended weatherproof area exceeds 50% of the existing weatherproof area, the extension is treated as a new building. The extended weatherproof area is measured as the cumulative area of any previous extensions plus the proposed extension.
FL4b	As for FL4a with the maximum percentage increase in extended weatherproof area to be: (a) 50% if the extension's floor level is less than one (1) metre below the 100 year flood level; (b) 25% if the extension's floor level is greater than two (2) metres below the 100 year flood level; or (c) pro-rata between 50% and 25% for floor levels from one (1) metre to two (2) metres below the 100 year flood level.
	BUILDING COMPONENTS
BC1	Buildings to have flood compatible material below the higher of (a) the minimum floor level or (b) the 1 in 100year flood level plus 0.5m.
	STRUCTURAL SOUNDNESS
SS1	No structural soundness requirements for the force of floodwater, debris & buoyancy
SS2	Engineers report to prove that structures subject to a flood up to the 100 year event can withstand the force of floodwater, debris & buoyancy.
SS3	Engineers report to prove that structures subject to a flood up to the 500 year event can withstand the force of floodwater, debris & buoyancy.
	FLOOD EFFECT
FE1	No action required
FE2	The flood impact of the development to be considered by Council, with Council having the right to request an engineer's report (see FE3 below)
FE3	Engineers report required to prove that the development will not result in adverse flood impact elsewhere
	EVACUATION/ACCESS
EA1	Council to provide information on flood evacuation strategy
EA2	Not used
EA3	Reliable access for pedestrians and transport required during the 100yr ARI event. Council to provide information on flood evacuation strategy
EA4a	Emergency service site - should have good access up to the PMF and preferably not cut-off from the main residential area(s). Council to evaluate suitablility of site in this respect.
EA4b	If site to be used during and after a flood emergency (see FL3b above), should have good access up to the PMF and preferably not cut-off from the main residential area(s).
	FLOOD AWARENESS
FA1	Not used
FA2	S149(2) Certificates to notify possible affectation by a flood in the Richmond River and/or one of its tributaries. The severity of flooding can be determined by comparison of surveyed levels of the site with predicted flood heights, and also the flood hazard.

MID-RICHMOND FLOOD PLANNING MATRIX

TABLE 1: RESIDENTIAL, COMMERCIAL AND INDUSTRIAL DEVELOPMENT WITHIN AN URBAN AREA

Controls	Development / Building Type	No Hazard	Flood Hazard Category					Additional Constraint[1]
			Rare Low Hazard[2]	Low Hazard	High Depth Hazard	High Isolation Hazard	High Floodway Hazard	Rare High Floodway Hazard[2]
Land Use Suitability & Fill Level	Existing Lot - including infill subdivision	N/A	SF1	SF1	SF1	SF1		SF1
	(this line not used)							
	Subdivision - en globo	N/A	SF2	SF2	SF2			
	Emergency Services Site (Hospitals, etc.)	N/A	SF3a	SF3a				
	Other Community Service Building (School, etc.)	N/A	SF3b	SF3b				
Floor Level	New Habitable Building	N/A	FL2c	FL2c	FL2c	FL2c		FL2c
	New Commercial or Industrial Building	N/A	FL2a	FL2a	FL2a	FL2a		FL2a
	New Emergency Service Building (Hospitals, etc.)	FL3a	FL3a	FL3a				
	New Other Community Service Building (School, etc.)	FL3b	FL3b	FL3b				
	New Ancillary Building (eg shed, carport)	N/A	FL1	FL1	FL1	FL1		FL1
	Building Extension	N/A	FL4a	FL4a	FL4b	FL4b		FL4b
	(this line not used)							
Building Components		N/A	BC1	BC1	BC1	BC1		BC1
Structural Soundness	Ancillary Building (eg. shed, carport)	N/A	SS1	SS1	SS1	SS1		SS2
	Other Building	N/A	SS1	SS1	SS2	SS2		SS3
Flood Effect	Existing Lot - including infill subdivision	N/A	FE1	FE2	FE2	FE2		FE2
	Subdivision - en globo	N/A	FE2	FE2	FE2			FE3
	New Ancillary Building (eg shed, carport)	N/A	FE1	FE2	FE2	FE2		FE2
	Building Extension	N/A	FE1	FE1	FE2	FE2		FE3
	Other Developments (road raising, etc)	N/A	FE1	FE2	FE2	FE2	FE3	FE3
Evacuation & Access	Existing Lot - including infill subdivision	N/A	EA1	EA1	EA1	EA1		EA1
	Subdivision - en globo	N/A	EA3	EA3	EA3			
	Emergency Service Site (Hospitals, etc.)	N/A	EA4a	EA4a				
	Other Community Service Site (Schools, etc.)	N/A	EA4b	EA4b				
Flood Awareness, etc		N/A	FA2	FA2	FA2	FA2	FA2	FA2

Note 1: In addition to being assigned one of the standard flood hazard categories, a site may be classified as a "Rare High Floodway Hazard". In this instance, the most stringent of the two controls is to be used. For example, if the site is classified as both "Low Hazard" and "Rare High Floodway Hazard", no community services buildings are permitted (because "unsuitable" is more stringent than SF3b)

Note 2: The extreme flood hazard categories (i.e. "Rare Low Hazard" and "Rare High Floodway Hazard") are applicable only to the 2D model region in Casino
An explanation of the criteria used to define the hazard categories is contained in the Casino Floodplain Management Study (WBM Oceanics Australia, 2001)

Note 3: Small-scale development implies development on rural land that is small relative to the width of the floodplain and is not part of a planned large-scale development.

Note 4: Weatherproof Area Definition - Enclosed areas excluding garages / carports / open verandahs

Control Measures

N/A	Controls Not Applicable
	Unsuitable Land Use - Not considered suitable for development
	LAND USE SUITABILITY & MINIMUM FILL LEVEL
SF1	Consider for development subject to the controls below. No minimum fill level required.
SF2	Consider for development subject to the controls below. For residential and commercial areas, the minimum fill level to be greater than or equal to the 100 year flood level. For industrial areas, the minimum fill level to be greater than or equal to the 10 year flood level.
SF3a	Consider for development subject to the controls below. Minimum fill level greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
SF3b	Consider for development subject to the controls below. Council to give consideration on the benefits of using the development during and after a flood emergency. If the site is to be used for a flood emergency, the minimum fill level should preferably be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
	MINIMUM FLOOR LEVEL
FL1	No minimum floor level required (Council to advise developer of flood risk and potential damage to building & contents. Flood levels available on request)
FL2a	Casino: All floor levels to be greater than or equal to the 100 year flood level Mid-Richmond: Minimum floor level greater than or equal to the 20 year flood level subject to access constraints and compatibility with adjoining land use. A storage area with a floor level equal to or greater than the 100 year flood level is to be provided for stock/equipment subject to water damage.
FL2b	For permissible uses other than residential, it is preferable to have all floor levels greater than or equal to the 100 year flood level subject to industry standards and individual site assessment.
FL2c	All habitable floor levels to be greater than or equal to the 100 year flood level plus 0.5m
FL3a	All floor levels to be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL3b	If practical, some or all floor levels to be greater than or equal to the PMF flood level, so that these buildings will be available for accommodation / storage during and after a flood emergency. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL4a	Habitable, commercial or industrial floor levels to be as close to the *minimum floor level* above as practical and not less than the floor level of the existing building being extended if the existing floor level is less than or equal to the minimum floor level. If the extended weatherproof area[4] exceeds 50% of the existing weatherproof area, the extension is treated as a new building. The extended weatherproof area is measured as the cumulative area of any previous extensions plus the proposed extension. If building is idenitified as being suitable for voluntary house raising scheme, Council to discuss potential house raising with owner.
FL4b	As for FL4a with the maximum percentage increase in extended weatherproof area[4] to be: (a) 50% if the extension's floor level is between the 100 year and 50 year flood levels; (b) 25% if the extension's floor level is less than or equal to the 10 year flood level; or (c) pro-rata between 50% and 25% for floor levels from the 50 year level to the 10 year level.
	BUILDING COMPONENTS
BC1	Buildings to have flood compatible material below the higher of (a) the minimum floor level or (b) the 1 in 100year flood level plus 0.5m.
	STRUCTURAL SOUNDNESS
SS1	No structural soundness requirements for the force of floodwater, debris & buoyancy. Must still comply with BCA requirements.
SS2	Engineers report to prove that structures subject to a flood up to the 100 year event can withstand the force of floodwater, debris & buoyancy.
SS3	Engineers report to prove that structures subject to a flood up to the 500 year event can withstand the force of floodwater, debris & buoyancy.
	FLOOD EFFECT
FE1	No action required
FE2	The flood impact of the development to be considered by Council, with Council having the right to request an engineer's report (see FE3 below)
FE3	Engineers report required to prove that the development will not result in adverse flood impact elsewhere
	EVACUATION/ACCESS
EA1	Council to provide information on flood evacuation strategy
EA2	Not used
EA3	Site specific Flood Evacuation Strategy be developed consistent with Council / SES overall Flood Evacuation Strategy.
EA4a	Emergency service site - should have good access up to the PMF and preferably not cut-off from the main residential area(s). Council to evaluate suitablility of site in this respect.
EA4b	If site to be used during and after a flood emergency (see FL3b above), should have good access up to the PMF and preferably not cut-off from the main residential area(s).
	FLOOD AWARENESS
FA1	Not used
FA2	S149(2) Certificates to notify possible affectation by a flood in the Richmond River and/or one of its tributaries. The severity of flooding can be determined by comparison of surveyed levels of the site with predicted flood heights, and also the flood hazard.

MID-RICHMOND FLOOD PLANNING MATRIX

TABLE 2: DEVELOPMENT IN RURAL AREAS

			Flood Hazard Category					Additional Constraint[1]
Controls	**Development / Building Type**	**No Hazard**	**Rare Low Hazard[2]**	**Low Hazard**	**High Depth Hazard**	**High Isolation Hazard**	**High Floodway Hazard**	**Rare High Floodway Hazard[2]**
Land Use Suitability & Fill Level	Habitable Building	N/A	SF1	SF1	SF1	SF1		SF1
	Ancillary Building (eg. shed)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	Other Developments (eg. levees, roads, dams, etc)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	Emergency Services Site (Hospitals, etc.)	N/A	SF3a	SF3a				
	Other Community Service Building (School, etc.)	N/A	SF3b	SF3b	SF3b			
Floor Level	New Habitable Building	N/A	FL2c	FL2c	FL2c	FL2c		FL2c
	(this line not used)							
	New Emergency Service Building (Hospitals, etc.)	FL3a	FL3a	FL3a				
	New Other Community Service Building (School, etc.)	FL3b	FL3b	FL3b				
	New Ancillary Building (eg shed, carport)	N/A	FL1	FL1	FL1	FL1	FL1	FL1
	Building Extension	N/A	FL4a	FL4a	FL4b	FL4b		FL4b
	New Rural Industry	N/A	FL2b	FL2b	FL2b	FL2b		FL2b
Building Components		N/A	BC1	BC1	BC1	BC1	BC1	BC1
Structural Soundness	Small-scale[3] Development (eg. shed, small dam)	N/A	SS1	SS1	SS1	SS1	SS2	SS3
	Large-scale Development (eg. levee, raised road)	N/A	SS1	SS1	SS2	SS2	SS2	SS3
Flood Effect	Small-scale[3] Development (eg. shed, small dam)	N/A	FE1	FE1	FE2	FE2	FE2	FE2
	Large-scale Development (eg. levee, raised road)	N/A	FE2	FE2	FE3	FE3	FE3	FE3
	(this line not used)							
	(this line not used)							
	(this line not used)							
Evacuation & Access	Habitable Building	N/A	EA1	EA1	EA1	EA1		EA1
	(this line not used)							
	Emergency Service Site (Hospitals, etc.)	N/A	EA4a	EA4a				
	Other Community Service Site (Schools, etc.)	N/A	EA4b	EA4b				
Flood Awareness, etc		N/A	FA2	FA2	FA2	FA2	FA2	FA2

Note 1: In addition to being assigned one of the standard flood hazard categories, a site may be classified as a "Rare High Floodway Hazard". In this instance, the most stringent of the two controls is to be used. For example, if the site is classified as both "Low Hazard" and "Rare High Floodway Hazard", no community services buildings are permitted (because "unsuitable" is more stringent than SF3b)

Note 2: The extreme flood hazard categories (i.e. "Rare Low Hazard" and "Rare High Floodway Hazard") are applicable only to the 2D model region in Casino
An explanation of the criteria used to define the hazard categories is contained in the Casino Floodplain Management Study (WBM Oceanics Australia, 2001)

Note 3: Small-scale development implies development on rural land that is small relative to the width of the floodplain and is not part of a planned large-scale development.

Note 4: Weatherproof Area Definition - Enclosed areas excluding garages / carports / open verandahs

Control Measures

N/A	Controls Not Applicable
	Unsuitable Land Use - Not considered suitable for development
	LAND USE SUITABILITY & MINIMUM FILL LEVEL
SF1	Consider for development subject to the controls below. No minimum fill level required.
SF2	Consider for development subject to the controls below. For residential and commercial areas, the minimum fill level to be greater than or equal to the 100 year flood level. For industrial areas, the minimum fill level to be greater than or equal to the 10 year flood level.
SF3a	Consider for development subject to the controls below. Minimum fill level greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
SF3b	Consider for development subject to the controls below. Council to give consideration on the benefits of using the development during and after a flood emergency. If the site is to be used for a flood emergency, the minimum fill level should preferably be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
	MINIMUM FLOOR LEVEL
FL1	No minimum floor level required (Council to advise developer of flood risk and potential damage to building & contents. Flood levels available on request)
FL2a	Casino: All floor levels to be greater than or equal to the 100 year flood level Mid-Richmond: Minimum floor level greater than or equal to the 20 year flood level subject to access constraints and compatibility with adjoining land use. A storage area with a floor level equal to or greater than the 100 year flood level is to be provided for stock/equipment subject to water damage.
FL2b	For permissible uses other than residential, it is preferable to have all floor levels greater than or equal to the 100 year flood level subject to industry standards and individual site assessment.
FL2c	All habitable floor levels to be greater than or equal to the 100 year flood level plus 0.5m
FL3a	All floor levels to be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL3b	If practical, some or all floor levels to be greater than or equal to the PMF flood level, so that these buildings will be available for accommodation / storage during and after a flood emergency. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL4a	Habitable, commercial or industrial floor levels to be as close to the *minimum floor level* above as practical and not less than the floor level of the existing building being extended if the existing floor level is less than or equal to the minimum floor level. If the extended weatherproof area[4] exceeds 50% of the existing weatherproof area, the extension is treated as a new building. The extended weatherproof area is measured as the cumulative area of any previous extensions plus the proposed extension. If
FL4b	As for FL4a with the maximum percentage increase in extended weatherproof area[4] to be: (a) 50% if the extension's floor level is between the 100 year and 50 year flood levels; (b) 25% if the extension's floor level is less than or equal to the 10 year flood level; or (c) pro-rata between 50% and 25% for floor levels from the 50 year level to the 10 year level.
	BUILDING COMPONENTS
BC1	Buildings to have flood compatible material below the higher of (a) the minimum floor level or (b) the 1 in 100year flood level plus 0.5m.
	STRUCTURAL SOUNDNESS
SS1	No structural soundness requirements for the force of floodwater, debris & buoyancy. Must still comply with BCA requirements.
SS2	Engineers report to prove that structures subject to a flood up to the 100 year event can withstand the force of floodwater, debris & buoyancy.
SS3	Engineers report to prove that structures subject to a flood up to the 500 year event can withstand the force of floodwater, debris & buoyancy.
	FLOOD EFFECT
FE1	No action required
FE2	The flood impact of the development to be considered by Council, with Council having the right to request an engineer's report (see FE3 below)
FE3	Engineers report required to prove that the development will not result in adverse flood impact elsewhere
	EVACUATION/ACCESS
EA1	Council to provide information on flood evacuation strategy
EA2	Not used
EA3	Site specific Flood Evacuation Strategy be developed consistent with Council / SES overall Flood Evacuation Strategy.
EA4a	Emergency service site - should have good access up to the PMF and preferably not cut-off from the main residential area(s). Council to evaluate suitablility of site in this respect.
EA4b	If site to be used during and after a flood emergency (see FL3b above), should have good access up to the PMF and preferably not cut-off from the main residential area(s).
	FLOOD AWARENESS
FA1	Not used
FA2	S149(2) Certificates to notify possible affectation by a flood in the Richmond River and/or one of its tributaries. The severity of flooding can be determined by comparison of surveyed levels of the site with predicted flood heights, and also the flood hazard.

MID-RICHMOND FLOOD PLANNING MATRIX

TABLE 3: OTHER

			Flood Hazard Category					Additional Constraint[1]
Controls	**Development / Building Type**	**No Hazard**	**Rare Low Hazard[2]**	**Low Hazard**	**High Depth Hazard**	**High Isolation Hazard**	**High Floodway Hazard**	**Rare High Floodway Hazard[2]**
Land Use Suitability & Fill Level	Non-Habitable Building (shed, toilets, shelter, etc)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	(this line not used)							
	Other Developments (eg. levees, roads, dams, etc)	N/A	SF1	SF1	SF1	SF1	SF1	SF1
	(this line not used)							
	(this line not used)							
Floor Level	(this line not used)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	(this line not used)							
	(this line not used)							
	(this line not used)							
	(this line not used)							
	(this line not used)							
	(this line not used)							
Building Components		N/A	BC1	BC1	BC1	BC1	BC1	BC1
Structural Soundness	Small-scale[3] Development (eg. shed, small dam)	N/A	SS1	SS1	SS1	SS1	SS2	SS3
	Large-scale Development (eg. levee, raised road)	N/A	SS1	SS1	SS2	SS2	SS2	SS3
Flood Effect	Small-scale Development (eg. shed, small dam)	N/A	FE1	FE1	FE2	FE2	FE2	FE2
	Large-scale Development (eg. levee, raised road)	N/A	FE2	FE2	FE3	FE3	FE3	FE3
	(this line not used)							
	(this line not used)							
	(this line not used)							
Evacuation & Access	Not Applicable							
	(this line not used)							
	(this line not used)							
	(this line not used)							
Flood Awaren	Not Applicable							

Note 1: **In addition to being assigned one of the standard flood hazard categories, a site may be classified as a "Rare High Floodway Hazard". In this instance, the most stringent of the two controls is to be used. For example, if the site is classified as both "Low Hazard" and "Rare High Floodway Hazard", no community services buildings are permitted (because "unsuitable" is more stringent than SF3b)**

Note 2: **The extreme flood hazard categories (i.e. "Rare Low Hazard" and "Rare High Floodway Hazard") are applicable only to the 2D model region in Casino. An explanation of the criteria used to define the hazard categories is contained in the Casino Floodplain Management Study (WBM Oceanics Australia, 2001)**

Note 3: **Small-scale development implies development on rural land that is small relative to the width of the floodplain and is not part of a planned large-scale development.**

Note 4: **Weatherproof Area Definition - Enclosed areas excluding garages / carports / open verandahs**

Control Measures

N/A	Controls Not Applicable
	Unsuitable Land Use - Not considered suitable for development
	LAND USE SUITABILITY & MINIMUM FILL LEVEL
SF1	Consider for development subject to the controls below. No minimum fill level required.
SF2	Consider for development subject to the controls below. For residential and commercial areas, the minimum fill level to be greater than or equal to the 100 year flood level. For industrial areas, the minimum fill level to be greater than or equal to the 10 year flood level.
SF3a	Consider for development subject to the controls below. Minimum fill level greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
SF3b	Consider for development subject to the controls below. Council to give consideration on the benefits of using the development during and after a flood emergency. If the site is to be used for a flood emergency, the minimum fill level should preferably be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
	MINIMUM FLOOR LEVEL
FL1	No minimum floor level required (Council to advise developer of flood risk and potential damage to building & contents. Flood levels available on request)
FL2a	Casino: All floor levels to be greater than or equal to the 100 year flood level Mid-Richmond: Minimum floor level greater than or equal to the 20 year flood level subject to access constraints and compatibility with adjoining land use. A storage area with a floor level equal to or greater than the 100 year flood level is to be provided for stock/equipment subject to water damage.
FL2b	For permissible uses other than residential, it is preferable to have all floor levels greater than or equal to the 100 year flood level subject to industry standards and individual site assessment.
FL2c	All habitable floor levels to be greater than or equal to the 100 year flood level plus 0.5m
FL3a	All floor levels to be greater than or equal to the PMF flood level. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL3b	If practical, some or all floor levels to be greater than or equal to the PMF flood level, so that these buildings will be available for accommodation / storage during and after a flood emergency. Mid-Richmond: If no site exists that can practically fulfill the above PMF requirement, the 500 year flood level plus 0.5m may substitute
FL4a	Habitable, commercial or industrial floor levels to be as close to the *minimum floor level* above as practical and not less than the floor level of the existing building being extended if the existing floor level is less than or equal to the minimum floor level. If the extended weatherproof area[4] exceeds 50% of the existing weatherproof area, the extension is treated as a new building. The extended weatherproof area is measured as the cumulative area of any previous extensions plus the proposed extension. If
FL4b	As for FL4a with the maximum percentage increase in extended weatherproof area[4] to be: (a) 50% if the extension's floor level is between the 100 year and 50 year flood levels; (b) 25% if the extension's floor level is less than or equal to the 10 year flood level; or (c) pro-rata between 50% and 25% for floor levels from the 50 year level to the 10 year level.
	BUILDING COMPONENTS
BC1	Buildings to have flood compatible material below the higher of (a) the minimum floor level or (b) the 1 in 100year flood level plus 0.5m.
	STRUCTURAL SOUNDNESS
SS1	No structural soundness requirements for the force of floodwater, debris & buoyancy. Must still comply with BCA requirements.
SS2	Engineers report to prove that structures subject to a flood up to the 100 year event can withstand the force of floodwater, debris & buoyancy.
SS3	Engineers report to prove that structures subject to a flood up to the 500 year event can withstand the force of floodwater, debris & buoyancy.
	FLOOD EFFECT
FE1	No action required
FE2	The flood impact of the development to be considered by Council, with Council having the right to request an engineer's report (see FE3 below)
FE3	Engineers report required to prove that the development will not result in adverse flood impact elsewhere
	EVACUATION/ACCESS
EA1	Council to provide information on flood evacuation strategy
EA2	Not used
EA3	Site specific Flood Evacuation Strategy be developed consistent with Council / SES overall Flood Evacuation Strategy.
EA4a	Emergency service site - should have good access up to the PMF and preferably not cut-off from the main residential area(s). Council to evaluate suitablility of site in this respect.
EA4b	If site to be used during and after a flood emergency (see FL3b above), should have good access up to the PMF and preferably not cut-off from the main residential area(s).
	FLOOD AWARENESS
FA1	Not used
FA2	S149(2) Certificates to notify possible affectation by a flood in the Richmond River and/or one of its tributaries.